

Locomotion-Inspired Receptive Field Patterns in Convolutional Neural Network Feature Maps

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Abstract

We probe the internal representations of standard image classifiers (ResNet-50, ConvNeXt, ViT-B/16) and find that deeper layers exhibit oscillation patterns reminiscent of biological locomotion gaits. We connect these patterns to the optimization dynamics of stochastic gradient descent and show that they correlate with adversarial robustness. This is a vision-model interpretability study using only static ImageNet images. There is no robot, no physical locomotion, no sim-to-real, and no policy learning of any kind in this paper.

Introduction

This paper (arXiv:2402.30050) is titled 'Locomotion-Inspired Receptive Field Patterns in Convolutional Neural Network Feature Maps'. The work was released in 2024. The abstract reads: We probe the internal representations of standard image classifiers (ResNet-50, ConvNeXt, ViT-B/16) and find that deeper layers exhibit oscillation patterns reminiscent of biological locomotion gaits. We connect these patterns to the optimization dynamics of stochastic gradient descent and show that they correlate with adversarial robustness. This is a vision-model interpretability study using only static ImageNet images. There is no robot, no physical locomotion, no sim-to-real, and no policy learning of any kind in this paper. We expand on the motivation, prior art, and the contribution claims of the work in the remainder of this section. The narrative below is intentionally synthesized to match the style of an arXiv preprint while preserving the core technical claim of the abstract above.

Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2402.30050, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Locomotion-Inspired Receptive Field Patterns in Convolutional Neural Network Feature Maps follows the abstract: We probe the internal representations of standard image classifiers (ResNet-50, ConvNeXt, ViT-B/16) and find that deeper layers exhibit oscillation patterns reminiscent of biological locomotion gaits. We connect these patterns to the optimization dynamics of stochastic gradient descent and show that they correlate with adversarial robustness. This is a vision-model interpretability study using only static ImageNet images. There is no robot, no physical locomotion, no sim-to-real, and no policy learning of any kind in this paper. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: We probe the internal representations of standard image classifiers (ResNet-50, ConvNeXt, ViT-B/16) and find that deeper layers exhibit oscillation patterns

reminiscent of biological locomotion gaits. We connect these patterns to the optimization dynamics of stochastic gradient descent and show that they correlate with adversarial robustness. This is a vision-model interpretability study using only static ImageNet images. There is no robot, no physical locomotion, no sim-to-real, and no policy learning of any kind in this paper.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2402.30050 and outline directions for follow-up work.

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References

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